Lecture 23

3/13/26
JRA

Discussing two SR techniques that won NP

PALM – stochastic

Cell w/ Filamentous structures w/ fluorophores attached

Assume fluorophores dark at first
PA GFP
Hit w/ weak activation light 405nm
Activate w/ light
Weak enough so only a few fluorophores are activated – non overlapping fluorophores

Collect data from the activated fluorophores, computer tabulates centers, fluorophores bleach, REPEAT again & again green pink blue

Repeat up to 10 000 x

Stochastic switching, coordinate readout, ultimately build up an image

Image via "pointalism"

There are some subtleties

Center is calculated
Need enough signal to determine center well

$$\text{uncertainty} = \frac{\text{PSF width}}{\sqrt{m}}$$

m = # of collected photons

ex) If PSF width ≈ 200 & $m = 400$

$$\text{Uncertainty} = \frac{200}{\sqrt{400}} = 10 \text{ nm}$$

400 = # of collected photons

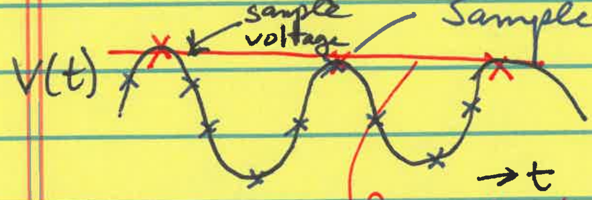
Dirty little secret of PALM

Pervasive issue in imaging

What's this?

<u>Sampling in Digital Imaging</u>

Sample → measure at discrete points

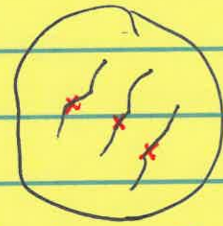


Sampling must be done frequently enough to reproduce the signal accurately

Bad: reconstruct sinusoidal signal as flat

This can cause big issues w/ PALM

PALM



Sample too sparsely
⇒ DOES NOT show filaments

To sample enough, need a LOT of repetitions
10000 x

Sampling Theorem tells how to sample correctly

EX | Let's say you want to achieve diffraction limited resolution w/ normal Fluorescence e.g. LSCM

$R = 200\,nm$

Want 200 nm resolution → won't get unless sample enough ⇒ sample size required $d = \frac{R}{2}$

For std Fluorescence $d = 100\,nm$

This is like pixel size. Cameras are adjusted.

$d = R/2$ sample size required $d = 100\,nm$

STED -

strengths - see blackboard photos

Processing (to find centers)

Can use cameras

targeted switching & coordinate readouts

Similar to <u>LSCM</u> - scanning technique

Spot is targeted to certain spot in cell

Know location Computer tabulates center position of spot r_i (ith spot)

Know position & location.

What's the advantage over LSCM?

STED scanning spot is ~ 10-fold smaller than an LSCM spot

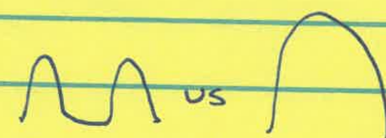
This is like painting w/ a finer brush

vs —

Laser spot Big - Fluorescece at same time ⇒ unresolvable

STED spot

Bright - Dark - Bright



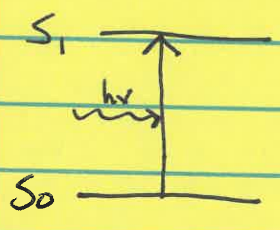
How do you make small spot?

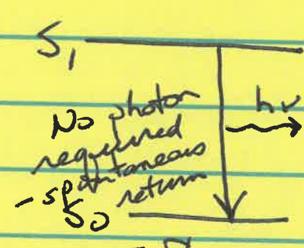
Stimulated Emission is the key to reducing spot size.

Stim absorption
Lrequires photon

spont emission

New kid
Stim emission



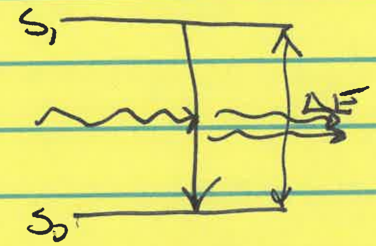


= Fluorescence

Needs photon to stimuftate absorption

MRI - everything aside stimulated

There is another form of radiation called stimulated emission
Predicted by Eistein LASER basis



Two identical output photons

Send in photon w/ energy matching energy between states – knock electron down out of excited state

Stim emission & STED

Focus now on STED application

Why 2 output photons?

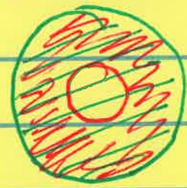
1 is energy of input photon

1 is energy loss going to ground state

2 output photons original & one from energy loss identical in every way

Engineering a small spot w/ stimulated emission. (1) Need two lasers.

(a) First laser excites fluorescence
Need some fluorescence to give signal



Big spot - size of LSCM spot
to excite fluorescence

Donut shaped spot to stimulate emission
to knock out of excited state
w/ stim emission in peripherary
only

Fluorophores in out depletted area no longer
Fluorescent - can't fluoresce

Leave a small unquenched spot to scan with

(2) Stimulated emission knocks fluorophores
out of excited state so they don't
fluoresce

Jargon: Quench Fluorescence in peripherary
of larger scanning spot
Need donut-shaped beam
using a D shaped beam

Do need two lasers
Green to excite
Red to quench

Stim energy photon is matched to emission transition
→ red shifted

Stokes shift says emission is lower E
⇒ red shifted

Use longer wavelength quenching beam

One more <u>subtlety</u>

This must occur super fast.

Stim emission when fluorophore in excited state

Molecule excited for ~ 1 ns
Hit while fluorophore still in excited state

Fluorophore must still be excited to be quenched by SE

Fluorescence lifetime is ~ 1 ns
Stimulated w/ donut in << 1 ns eg 100 psec

Requires 2 lasers, really fast

STED advantages

① Much faster than PALM
can do live specimen imaging

MOVE

② No image processing

③ Theoretically unlimited resolution
~20 nm

④

Weaknesses

① 2 lasers for every fluorophore   Hard for multiple fluorophores

② Very high light intensities < Photo B / Photo T

③ Tricky set ups

Microscopy concept Monday